

AI Model Recommendation

Vehicle Detection & Classification — Speed and Accuracy Options

Prepared: 2026-08-18

Summary

The current system already uses a strong, purpose-built AI model for this task: a YOLOv8 object-detection model trained specifically on Bangladeshi road traffic (the BNVD dataset), which recognizes region-specific vehicles — rickshaw, CNG/auto-rickshaw, easy-bike, and others — that generic international models do not. The main constraint on speed today is not the model itself, but the hardware it runs on: a standard laptop CPU. The recommendation below is to keep the same model and improve the hardware it runs on, rather than switching to a different AI model.

Recommendation

Run the existing model on a GPU (graphics processing unit) instead of a CPU. GPUs are 10–30× faster than CPUs for this kind of AI workload. This alone is expected to bring a 1-hour video down from multiple hours of processing time to roughly 10–30 minutes, without changing what the system detects or how accurately.

Two ways to get GPU access:

- **Local GPU** — if the client's own computer has a dedicated NVIDIA graphics card, the same software already built can use it directly, at no extra recurring cost.
- **Rented cloud GPU** — if no local GPU is available, a cloud GPU can be rented by the hour (typically \$0.20–\$1.00/hour) from providers such as RunPod, Lambda Labs, or AWS. The same model and detection logic runs unchanged, just on rented hardware.

Options Considered

Option	Speed (1-hr video)	Rickshaw/CNG detection	Ongoing cost	Verdict
Current setup (CPU only)	Several hours	Yes	None	Too slow for large videos
Same model, local GPU	~10–30 minutes	Yes	None (one-time hardware)	Recommended if GPU available
Same model, rented cloud GPU	~10–30 minutes	Yes	~\$0.20–\$1/hr of processing	Recommended if no local GPU
Generic cloud vision API (e.g. Google, AWS)	~10–30 minutes	No — generic vehicle types only	Per-frame or per-minute fees	Not recommended: loses regional accuracy
Newer YOLO version (v10/v11), same hardware	Modest improvement	Only if retrained on the same dataset	None	Worth trying, smaller impact

Why Not a Generic Cloud AI API?

General-purpose cloud vision services (Google Cloud, AWS Rekognition, and similar) can detect common vehicle types — cars, buses, trucks, motorcycles — quickly and without any local hardware. However, none of them recognize Bangladesh-specific vehicle types such as rickshaws, CNGs, or easy-bikes, which are central to the vehicle categories this project needs to report on. Using one of these services would mean losing that accuracy in exchange for convenience, and would introduce a recurring per-use cost. The current model already solves the accuracy problem; the recommended path solves the speed problem without giving that up.

Next Step

Confirm whether the client's computer has a dedicated NVIDIA graphics card. If yes, the existing software can be configured to use it directly at no additional cost. If no, the recommended path is a short trial on a rented cloud GPU to confirm real-world processing time on an actual 1-hour video before committing to a longer-term setup.